Answer **all** the questions in this Section in the spaces provided

**1** **(a)** Make estimates of the following quantities.

**(i)** the thickness of a sheet of A4 paper

thickness = ..... mm [1]

**(ii)** the mass of a sheet of A4 paper

mass = ..... g [1]

**(b)** The distance from the Earth to the Sun is 0.15 Tm.

Calculate the time in minutes for light to travel from the Sun to the Earth.

time = ..... minutes [2]

**(c)** The time $T$ for a satellie to orbit the Earth is given by

$$T = \sqrt{\frac{KR^3}{M}}$$

where $R$ is the distance of the satellite from the centre of the Earth, $M$ is the mass of the Earth and $K$ is a constant.

**(i)** Determine the SI base units of $K$.

SI base units of $K$ = ..... [2]

For Examiner's Use

**(ii)** Data for a particular satellite are given in Fig. 1.1

| quantity | measurement | uncertainty |
|---|---|---|
| $T$ | $8.64 \times 10^4$ s | ± 0.50% |
| $R$ | $4.23 \times 10^7$ m | ± 1.0% |
| $M$ | $6.0 \times 10^{24}$ kg | ± 2.0% |

**Fig. 1.1**

Express $K$ with its associated uncertainty in SI units.

$K$ = ……± …… SI units [3]

**(iii)** State the quantity which contributes the largest uncertainty in the value of $K$.

……

…….. [1]